Communications and Networks (Spring 2022)

Problem Set 2

Issued: Mar. 14 Due: Mar. 27

Readings:
Nguyen - A First Course in Digital Communications 4.1.1, 4.3.1-4.3.4
Proakis - Digital Signal Processing 6.4.1

1. The frequency spectrum of a low-pass signal $m(t)$ is

$$M(f) = \begin{cases} 1 - \frac{|f|}{200}, & |f| < 200\text{Hz} \\ 0, & \text{else.} \end{cases}$$

   a) We first sample $m(t)$ with a periodic sequence of unit impulse functions at sampling rate $f_s = 300$Hz. Draw the frequency spectrum $M_s(f)$ of the sampled signal $m_s(t)$.

   b) If the sampling rate is increased to $f_s = 400$Hz, please draw $M_s(f)$ again.

2. In this problem, we examine the aliasing effect in the time domain.

   a) For a sinusoidal signal $m^{(1)}(t) = \cos(200\pi t)$, we conduct ideal sampling at $f_s = 300$Hz. Draw the sampled signal $m_s^{(1)}(t)$ along with $m^{(1)}(t)$. Also draw the frequency spectrum $M_s^{(1)}(f)$ of the sampled signal.

   b) For another sinusoidal signal $m^{(2)}(t) = \cos(400\pi t)$, the same sampling process is conducted. Draw $m_s^{(2)}(t)$ along with $m^{(2)}(t)$. Also draw the spectrum $M_s^{(2)}(f)$.

   c) Compare $m_s^{(1)}(t)$ and $m_s^{(2)}(t)$ and comment. What signal $m_r(t)$ do we actually reconstruct after applying a low-pass filter at the receiver?

3. A sinusoidal signal is uniformly quantized. Suppose the signal fits exactly inside the quantization range of the quantizer, and the dynamic range, i.e. the ratio of the largest value and the smallest value, is 40dB. How many bits are required at least for $\text{SNR}_q \geq 30$dB over the dynamic range? (Hint: the normalized effective value D of the sinusoidal wave is $1/\sqrt{2}$)

4. We are going to conduct the optimal quantization on a Gaussian random variable $X \sim \mathcal{N}(\eta, \sigma^2)$ with only one bit.

   a) Find the optimal decision level and representation points.

   b) Calculate the expected quantization noise, measured by mean squared error(MSE).

5. In this problem, we apply PCM (Pulse Code Modulation) in a voice communication system. It is widely known that human voice has a frequency range of approximately 0.3 to 4 kHz. Therefore in voice communication systems we only care about low-pass signals **below 4kHz**.

a) What is the minimum sampling rate f_s for perfect reconstruction below 4kHz?

b) In real-world voice signal processing, the total frequency band is divided into sub-bands for separate encoding. In specific, we consider 4 sub-bands, i.e. 0-1kHz, 1-2kHz, 2-3kHz, 3-4kHz. Determine the minimum sampling rate for each of these sub-bands. What is the total sampling rate?

c) Given the 8-bit coding protocol below, with input level -314, compute the PCM code, representation point and noise.

Segment	Segment code			Start Point	Sub-segment code				Quantization interval
	M_2	M_3	M_4		M_5	M_6	M_7	M_8	
0	0	0	0	0	16	8	4	2	2
1	0	0	1	32	16	8	4	2	2
2	0	1	0	64	32	16	8	4	4
3	0	1	1	128	64	32	16	8	8
4	1	0	0	256	128	64	32	16	16
5	1	0	1	512	256	128	64	32	32
6	1	1	0	1024	512	256	128	64	64
7	1	1	1	2048	1024	512	256	128	128

d) What is the total bit-rate of this voice communication system? Express the answer in kbps (kilo-bits per second).